30/11/2025

A

Aakash
Medical | IIT-JEE | Foundations

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MM : 720

OYM_NBTS PRACTICES TEST_30.11.25

Time : 180 Min.

PHYSICS

1. Minimum number of NOR gate required to make AND gate is

(1) 1

(2) 2

(3) 3

(4) 4

2. Two nuclei have their mass numbers in the ratio 1 : 8. The ratio of their nuclear densities would be

(1) 1 : 2

(2) $1 : 2\sqrt{2}$

(3) 1 : 1

(4) 1 : 8

3. In the nth state, energy of electron in hydrogen atom is $E_n = -\frac{13.6}{n^2}$ eV. The potential energy of electron in first excited state will be

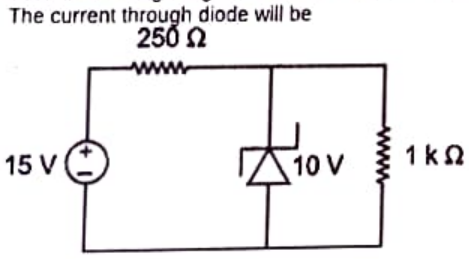
(1) –27.2 eV

(2) –3.4 eV

(3) –3.2 eV

(4) –6.8 eV

4. A zener diode, having breakdown voltage equal to 10 V is used in a voltage regulator circuit as shown in the figure. The current through diode will be



(1) 40 mA

(2) 20 mA

(3) 10 mA

(4) 5 mA

5. What will be the respective inputs *A* and *B* for the Boolean expression $(\overline{A+B})\cdot(\overline{A\cdot B}) = 1$?

(1) 0, 0

(2) 0, 1

(3) 1, 0

(4) 1, 1

6. Suppose a pure Silicon (Si) crystal has 6×10^{27} atoms m^{-3}. It is doped by 1 ppm concentration of pentavalent impurity. Calculate number of holes [Given that $n_i = 1.2 \times 10^{15}\ m^{-3}$.]

(1) $2.4 \times 10^{9}\ m^{-3}$

(2) $2.4 \times 10^{8}\ m^{-3}$

(3) $4.8 \times 10^{8}\ m^{-3}$

(4) $4.8 \times 10^{9}\ m^{-3}$

7. On the basis of Bohr's atomic model the radius of 3rd orbit of any atom is

(1) Three times of radius of first orbit

(2) Nine times of radius of first orbit

(3) One third of radius of first orbit

(4) Equal to the radius of first orbit

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8. The ratio of the radii of the nuclei of $_{13}Al^{27}$ and $_{52}Te^{125}$ is

(1) 1 : 4

(2) 3 : 5

(3) 9 : 25

(4) 27 : 125

9. The binding energy per nucleon of nuclei $^{56}_{26}Fe$ from the following data, will be

$m\left(^{56}_{26}Fe\right) = 55.934939$ u

$m_n = 1.008665$u

$m_H = 1.007825$ u

1 u = 931.5 MeV/c^2

(1) 20 MeV

(2) 8.79 MeV

(3) 15.65 MeV

(4) 2.12 MeV

10. LED is a p-n junction diode which is

(1) Heavily doped and forward biased

(2) Lightly doped and forward biased

(3) Heavily doped and reverse biased

(4) Lightly doped and reverse biased

11. The forward bias diode is

(1) 

(2) 

(3) 

(4) 

12. A : Number of possible absorption lines is always less than number of possible emission lines.
R : For n levels, the number of possible emission lines are $\frac{n(n-1)}{2}$ and the number of possible absorption lines are $(n-1)$.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

13. Binding energies per nucleons of nuclei 'A' and 'B' are 6.8 MeV and 5 MeV respectively. Then correct option is

(1) B is more stable

(2) A is more stable

(3) A & B are equally stable

(4) Stability does not depend on binding energy per nucleon

14. The nuclei of which of the following pairs of nuclei are isotone?

(1) ${}_{34}Sc^{74}, {}_{31}Ca^{71}$

(2) ${}_{32}Mo^{92}, {}_{40}Zr^{92}$

(3) ${}_{38}Sr^{81}, {}_{38}Sr^{86}$

(4) ${}_{20}Cs^{40}, {}_{16}S^{32}$

15. A : Nuclear forces are saturated forces.
R : Nuclear forces are long range force.

(1) Assertion is true statement but Reason is false

(2) Both Assertion and Reason are false statements

(3) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(4) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

16. If n is the principal quantum number, then de-Broglie wavelength associated with n^{th} orbit of Bohr's model is proportional to

(1) n^0

(2) $\frac{1}{n}$

(3) n

(4) n^2

17. For a semiconductor, choose the correct statement

(1) $n_e >> n_h$ for p-type semiconductor

(2) $n_e >> n_h$ for intrinsic semiconductor

(3) $n_e = n_h$ for n-type semiconductor

(4) $n_e << n_h$ for p-type semiconductor

18. The radiation emitted when an electron jumps from $n = 3$ to $n = 2$ orbit in hydrogen atom falls on a metal to produce photoelectrons. If work function of metal is 0.8 eV, stopping potential will be

(1) 1.09 V

(2) 2.09 V

(3) 3.09 V

(4) 5.09 V

19. Longest wavelength of balmer series of H–atom is 6563 Å. Longest wavelength of Lyman series then, will be

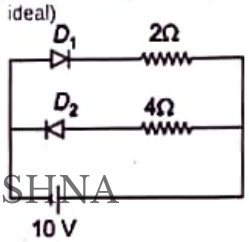
(1) 1215 Å

(2) 2245 Å

(3) 6000 Å

(4) 4560 Å

20. Current through 2 Ω resistor in the circuit is (All diodes are ideal)



(1) 5 A

(2) 2.5 A

(3) Zero

(4) 1.75 A

21. If a hydrogen atom in ground state absorb 12.09 eV of energy, then angular momentum of electron in highest excited state is

(1) $\frac{3h}{2\pi}$

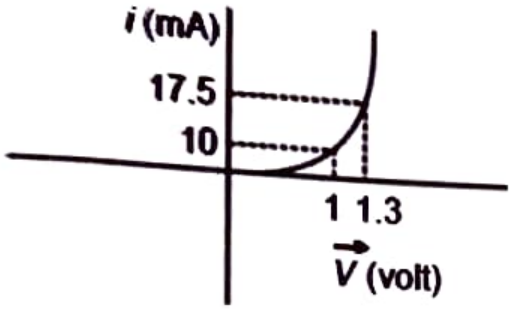
(2) $\frac{h}{2\pi}$

(3) $\frac{2h}{2\pi}$

(4) $\frac{4h}{2\pi}$

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22. The *I-V* characteristic of a *p-n* junction diode is shown. Find the approximate dynamic resistance of the *p-n* junction when a forward bias of 1 volt is applied.



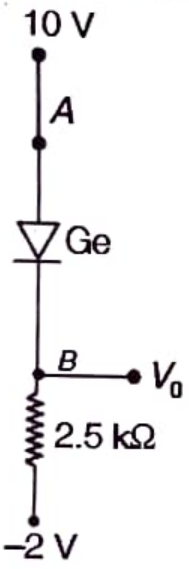
(1) 20 Ω

(2) 40 Ω

(3) 50 Ω

(4) 60 Ω

23. For the circuit as shown in figure the value of voltage V_0 is



(1) 12 V

(2) 9.7 V

(3) 11.7 V

(4) 11 V

24. Pure Si at 350 K has equal electron (n_e) and hole (n_h) concentration of 3×10^{16} per m^3. Doping by indium increases n_h to 9×10^{22} per m^3. Calculate n_e in the doped silicon

(1) 1×10^{10} per m^3

(2) 2×10^{10} per m^3

(3) 3×10^{10} per m^3

(4) 4×10^{10} per m^3

25. The ratio of angular speed and tangential speed of electron in n^{th} orbit of hydrogen atom is

(1) Inversely proportional to n^2

(2) Directly proportional to n^2

(3) Inversely proportional to n^4

(4) Independent of n

26. A : In Boolean algebra, $\overline{A+B} = \bar{A} + \bar{B}$.
R : In Boolean algebra, $\overline{A \cdot B} = \bar{A} \cdot \bar{B}$.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

27. The difference between $(n+1)^{th}$ Bohr radius and n^{th} Bohr radius is equal to the $(n-1)^{th}$ Bohr radius. The value of n is

(1) 3

(2) 4

(3) 5

(4) 6

28. If the mirror formula is given by $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, then the error in the measurement of focal length is given by

(1) $\Delta f = f\left[\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}\right]$

(2) $\Delta f = 2f\left[\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}\right]$

(3) $\Delta f = f^2\left[\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}\right]$

(4) $\Delta f = f^2\left[\frac{\Delta u}{u^2} - \frac{\Delta v}{v^2}\right]$

29. The viscous force acting between two layers of a liquid is given by $\frac{F}{A} = -\eta\frac{dv}{dz}$. The term $\frac{dv}{dz}$ may be called (where symbols have their usual meaning)

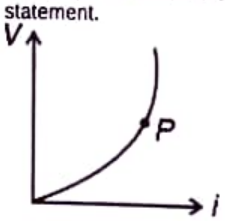
(1) Shear strain

(2) Shear strain rate

(3) Velocity gradient

(4) Both (2) & (3)

30. *V –i* graph for a circuit is shown below. Choose the correct statement.



(1) Resistance *R* for the wire used in circuit is constant

(2) The current through the circuit is directly proportional to potential difference across it

(3) The circuit do not follow ohm's law and is non-ohmic

(4) $V = iR$ is not applicable for a general point *P* on the graph

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31. The main scale of a vernier caliper is calibrated in mm and 9 divisions of main scale are equal in length to 10 divisions of vernier scale. In measuring the diameter of a cylinder by this instrument. The main scale reads 26 divisions and 5th division of vernier scale coincides with a main scale division. Radius of cylinder will be

(1) 2.51 cm

(2) 2.65 cm

(3) 1.32 cm

(4) 3.52 cm

32. A galvanometer has full deflection current of 10 mA. If there are total 50 division in the galvanometer scale, then figure of merit of galvanometer is

(1) 4 mA/div

(2) 40 mA/div

(3) 400 μA/div

(4) 40 μA/div

33. In determination of Young's modulus $\left(Y = \frac{4MLg}{\pi l d^2}\right)$ by using Searle's method, a wire of length L = 2 m and diameter = 1 mm is used. For a load M = 4 kg an extension l = 0.5 mm in length is observed. Quantities d and l are measured using screw gauge and micrometre, respectively. They have same pitch of 1 mm. The number of divisions on their circular scale is 100. The percentage relative error in the measurement of Y is

(1) 2%

(2) 4%

(3) 6%

(4) 8%

34. The energy equivalent of 1 kg of substance is

(1) 10^{16} J

(2) 10^{16} MeV

(3) 9×10^{16} J

(4) 9×10^{16} MeV

35. A screw gauge gives the following readings when used to measure the diameter of a wire
Main scale reading : 1 mm
Circular scale reading : 48 divisions
Given that 2 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is

(1) 1.96 mm

(2) 1.96 cm

(3) 0.96 mm

(4) 0.96 cm

36. In a vernier calliper, N division of vernier scale coincide with $(N - 1)$ division of main scale. The least count of the instrument (in cm) is
(1 division of main scale is 1 mm)

(1) N

(2) $N - 1$

(3) $\frac{1}{10N}$

(4) $\frac{1}{N-1}$

37. For a moving coil galvanometer which one of the following statements is correct?

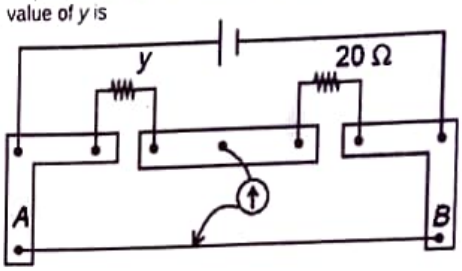
(1) It is more stable if it is more sensitive

(2) It is less stable if it is more sensitive

(3) Its stability and sensitivity are independent of each other

(4) Up to certain limit sensitivity increases stability but after that decreases stability

38. A meter bridge is set-up as shown, to determine an unknown resistance y using a standard resistor of 20 Ω. The galvanometer shows null point when tapping-key is at 42 cm mark. The end corrections are 1 cm and 2 cm respectively for the ends A and B, then the determined value of y is



(1) 10.2 Ω

(2) 12.3 Ω

(3) 14.3 Ω

(4) 20.3 Ω

39. In half deflection method, if value of resistance of galvanometer (G) is 100 Ω and resistance (R) is 1 kΩ, then the value of shunt resistance (S) is such that

(1) $S = G$

(2) $S = R$

(3) $S > G$

(4) $S < G$

40. A metre scale is balanced by placing the wedge at centre. If a 2 kg mass is hung at 15 cm mark and a 4 kg mass is hang at the 75 cm mark then to which mark on the metre scale the fulcrum should be shifted to balance the scale? (Neglect mass of meter scale)?

(1) 40 cm

(2) 55 cm

(3) 50 cm

(4) 60 cm

41. Avalanche breakdown in a PN junction diode is due to

(1) Sudden shift to fermi level

(2) Increase in the width of Forbidden gap

(3) Sudden increase of impurity concentration

(4) Cumulative effect of increased electron collision and creation of added electron hole pair

42. As the quantum number increases, the difference of energy between consecutive energy levels:

(1) Decreases

(2) Increases

(3) Remains same

(4) First decreases then increases

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43. A glass tube of radius R is dipped vertically in water. The water rises up to height h in the capillary. If the tube is inclined at an angle of 60° with vertical then the length of capillary occupied by water is

(1) $h\cos 60°$

(2) $\frac{h}{\cos 60°}$

(3) $h\cos 30°$

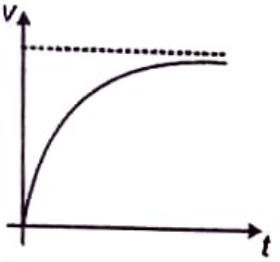
(4) $\frac{h}{\cos 30°}$

44. A small metal sphere is dropped on water surface from zero height. The velocity (v) of the sphere varies with time as

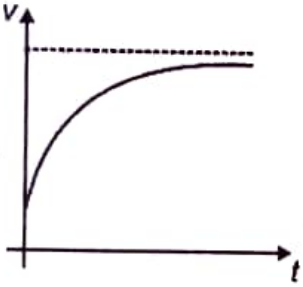
(1)



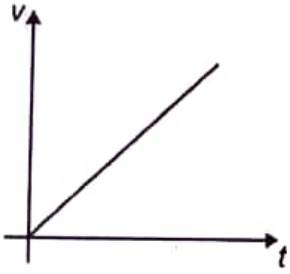
(2)



(3)



(4)



45. A small red mark on a table top is viewed from a distance of 20 cm. It appears to be raised from the same point when viewed through a 12 cm thick glass slab of refractive index 1.5 held parallel to the table. The distance through which it appears to be raised is

(1) 12 cm

(2) 6 cm

(3) 4 cm

(4) 2 cm

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CHEMISTRY

46. For the complex conversion of D-glucose (1 eq) into the corresponding osazone, the minimum number of equivalents of phenyl hydrazine required is

(1) 2

(2) 3

(3) 4

(4) 5

47. A : Aniline does not undergo Friedel Craft reaction.
R : $-NH_2$ group in Aniline is deactivating group towards electrophilic substitution.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

48. Nitrobenzene is converted into aniline in weakly acidic medium through electro reduction. The n-factor of nitrobenzene in the given conversion is

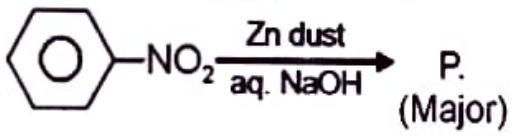
(1) 3

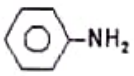
(2) 6

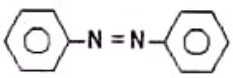
(3) 4

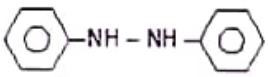
(4) 2

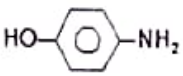
49. The product, P formed in the given reaction is



(1) 

(2) 

(3) 

(4) 

50. Incorrect statement among the following is

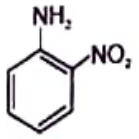
(1) Amylose is water soluble component of starch

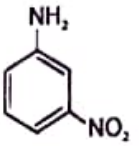
(2) Amylose is long unbranched chain having α-D-(+) glucose units

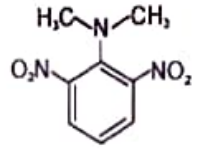
(3) Amylopectin is branched chain polymer of α-D-glucose units

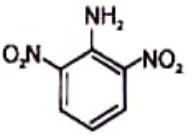
(4) Amylopectin constitutes about 15-20% of starch

51. pK_b will be minimum for which molecule?

(1) 

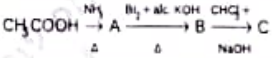
(2) 

(3) 

(4) 

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52. Consider the following sequence of reactions:



Which of the following statements is incorrect with respect to compound C?

(1) Foul smelling compound

(2) Contains same number of carbon atoms as that of A

(3) Reduction of compound C with $LiAlH_4$ gives primary amines

(4) Can show functional isomerism

53. From 44 mg CO_2, 10^{20} molecules are removed calculate the moles of CO_2 left (Use $N_A = 6 \times 10^{23}$)

(1) 4.25×10^{-4}

(2) 8.3×10^{-4}

(3) 1.6×10^{-2}

(4) 4.4×10^{-2}

54. The ratio of the frequency of revolution of electron in the first excited state of He^+ and the third excited state of hydrogen is

(1) 1 : 9

(2) 32 : 27

(3) 32 : 1

(4) 27 : 32

55. When a metal is exposed to light of wavelength (λ), photoelectrons with maximum kinetic energy 2 eV are ejected from metal surface. On exposing the same metal with light of wavelength $\left(\frac{\lambda}{4}\right)$, the ejected photoelectrons have maximum kinetic energy 10 eV. The work function of metal is

(1) $\frac{2}{3}$ eV

(2) $\frac{1}{3}$ eV

(3) 2 eV

(4) 3 eV

56. If the wavelength of the first line of He^+ ion in Lyman series is x Å, then the wavelength of the first line of Li^{2+} in Balmer series will be

(1) $\frac{2x}{9}$ Å

(2) $\frac{12x}{5}$ Å

(3) 3 x Å

(4) $\frac{x}{9}$ Å

57. If the mass percentage of magnesium in a biomolecule is 0.04%, then the minimum molecular weight of biomolecule is

(1) 6000

(2) 60000

(3) 30000

(4) 12000

58. The mass of 80% pure H_2SO_4 required to neutralise 106 g of Na_2CO_3 is

(1) 78.4 g

(2) 122.5 g

(3) 98 g

(4) 80 g

59. The dissociation of O_2 molecule into two O atoms occurs by absorbing light of wavelength equal to 1750 Å. Assuming each O_2 molecule absorbs one quantum of radiation, the kinetic energy of one oxygen atom will be (Given : Dissociation energy of O_2 = 8.269×10^{-19} J/molecule)

(1) 3.01×10^{-19} J

(2) 1.545×10^{-19} J

(3) 6.182×10^{-19} J

(4) 8.269×10^{-19} J

60. The crystalline salt $Na_2CO_3.xH_2O$ on heating loses 63% of its mass and becomes anhydrous. The formula of the compound is

(1) $Na_2CO_3.2H_2O$

(2) $Na_2CO_3.10H_2O$

(3) $Na_2CO_3.5H_2O$

(4) $Na_2CO_3.8H_2O$

61. The total number of electrons in chromium atom with n + l = 3, is

(1) 8

(2) 2

(3) 10

(4) 6

62. 200 ml of 3 M NaOH solution is diluted by 400 ml water. The molarity of the resulting solution is

(1) 1 M

(2) 2 M

(3) 1.5 M

(4) 2.25 M

63. The empirical formula of monosodium glutamate (MSG) a food-flavor enhancer is $C_5H_8O_4$ NNa. If its vapour density is 84.5 then number of atoms in one molecule of MSG is

(1) 17

(2) 20

(3) 18

(4) 19

64. If the molality of methanol in an aqueous solution is 6 then the mass percentage of methanol in the solution will be

(1) 24.2%

(2) 16.1%

(3) 8.4%

(4) 31.5%

65. Which among the following species contains highest number of unpaired electrons?

(1) Cr^{2+}

(2) Co^{2+}

(3) Fe^{2+}

(4) Mn^{2+}

66. Number of electrons present in a manganese atom having quantum numbers l = 1, m = 0 and s = + $\frac{1}{2}$ is

(1) 12

(2) 6

(3) 4

(4) 2

67. A : Energy is absorbed for the transition of e^- from 4s to 3d level in hydrogen atom.
R : Energy of electron in all atoms depends on n + l value.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

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COOH

68. If x g of a metal chloride contains y g of metal then equivalent weight of metal will be

(1) $\frac{35.5x}{x-y}$

(2) $\frac{35.5y}{x-y}$

(3) $\frac{35.5x}{y}$

(4) $\frac{x}{35.5y}$

69. Total possible exchange for a f^6 configuration is

(1) 6 electrons

(2) 12 electrons

(3) 15 electrons

(4) 28 electrons

70. What is the mass of the precipitate formed when 50 mL of 20.8% (w/V) solution of $BaCl_2$ is mixed with 50 mL of 14.2 % (w/V) Na_2SO_4 solution? (g atomic mass of barium = 137 g mol^{-1})

(1) 10 g

(2) 11.65 g

(3) 23.3 g

(4) 5.825 g

71. Number of Primary amines of the formula $C_4H_{11}N$ is

(1) 1

(2) 2

(3) 3

(4) 4

72. The correct increasing order of bond dissociation enthalpy of halogens is

(1) $I_2 < Br_2 < Cl_2 < F_2$

(2) $I_2 < F_2 < Br_2 < Cl_2$

(3) $F_2 < I_2 < Br_2 < Cl_2$

(4) $Br_2 < F_2 < I_2 < Cl_2$

73. Given below are two statements one is assertion (A) the other is reason (R).
Assertion (A) : SO_2 is anhydride of sulphurous acid.
Reason (R) : When SO_2 is passed through lime water, it becomes milky.
In the light of above statements, choose the **correct** answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

74. Which among the following nitrogeneous base is absent in RNA?

(1) Uracil

(2) Thymine

(3) Cytosine

(4) Guanine

75. Incorrect statement among the following is

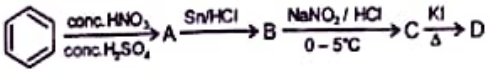
(1) Proline is a non-essential amino acid

(2) Glycine is an optically inactive molecule

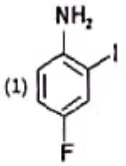
(3) Myosin is fibrous protein

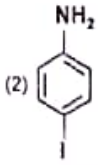
(4) Insulin has fibre-like structure

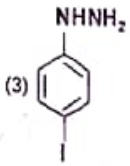
76. Consider the following reaction



Product D is

(1) 

(2) 

(3) 

(4) 

77. pK_1 and pK_2 of an amino acid are 2.3 and 9.7, then isoelectric point of the amino acid is

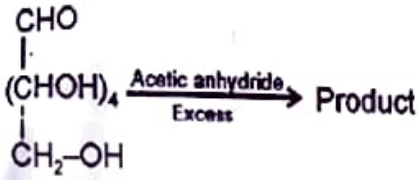
(1) 1.15

(2) 2.7

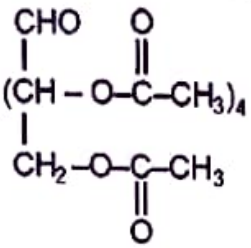
(3) 4.85

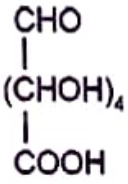
(4) 6.0

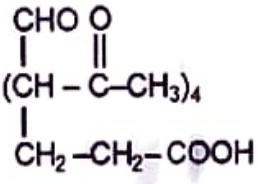
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For Preview Only
Do not print, save paper

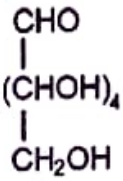
78. 

The product of the above reaction would be

(1) 

(2) 

(3) 

(4) 

79. The following reaction is performed for the detection of NO_3^- ions

$FeSO_4 + NO \rightarrow$ Product

Choose the correct statement

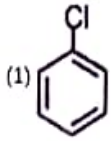
(1) Colour of the complex compound formed is violet

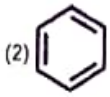
(2) Oxidation number of 'Fe' in complex is +3

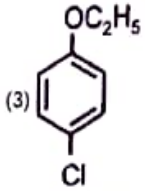
(3) The complex formed is $[Fe(H_2O)_5NO]^{2+}$

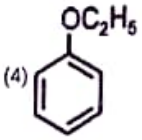
(4) Colour of the complex product is yellow

80. When benzene diazonium chloride is treated with ethanol then the product obtained is

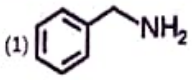
(1) 

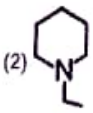
(2) 

(3) 

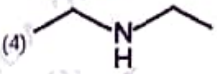
(4) 

81. Which compound does not react with Hinsberg's reagent?

(1) 

(2) 

(3) 

(4) 

82. In the following reactions

Acetanilide $\xrightarrow[CH_3COOH]{Br_2}$ C $\left(\text{major}\right) \xrightarrow{H^+}$ D.D is

(1) 2-Bromoaniline

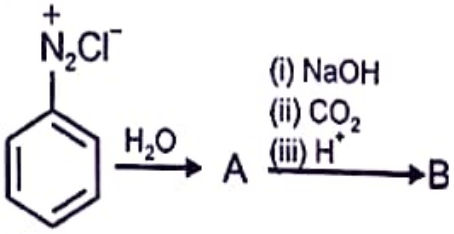
(2) 3-Bromoaniline

(3) 4-Bromoaniline

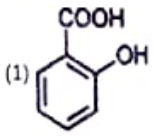
(4) 2, 4, 6-Tribromoaniline

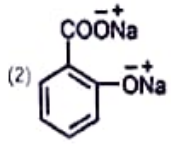
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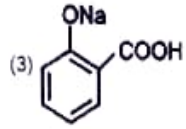
83.

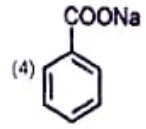


B is

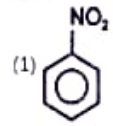
(1) 

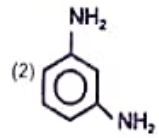
(2) 

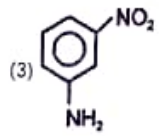
(3) 

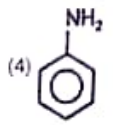
(4) 

84. Reaction of m-dinitrobenzene with ammonium sulphide yields

(1) 

(2) 

(3) 

(4) 

85. Consider the following statements

Statement-I: Secondary structure of protein arises due to the regular folding of the backbone of the polypeptide chain due to H-bonding between-CO- and –NH–groups of the peptide bond.

Statement-II: During denaturation of protein 2° and 3° structures are destroyed but 1° structure remains intact.

In the light of above statements, choose the correct answer from the options given below

(1) Statement I is incorrect but statement II is true

(2) Statement I is correct but statement II is false

(3) Both statements I and II are false

(4) Both statements I and II are true

86. Group reagent for precipitating the ion Ba^{2+} is

(1) $H_2S + NH_4OH$

(2) $(NH_4)_2CO_3 + NH_4OH$

(3) Dil. HCl

(4) $NH_4OH + NH_4Cl$

87. Methyl orange indicator is used in the acid-base titration whose pH range lies between____.

(1) 3.1-4.5

(2) 1-3

(3) 7-11

(4) 11-14

88. The metal ion that imparts apple green colour to the flame test is

(1) Cu^{2+}

(2) Sr^{2+}

(3) Ba^{2+}

(4) Ca^{2+}

89. In the detection of 'nitrogen' using Lassaigne test, the formation of a Prussian blue precipitate is due to the formation of

(1) $Fe(SCN)_3$

(2) $Fe_4[Fe(CN)_6]_3$

(3) $Fe_2[Fe(CN)_6]_3$

(4) $K_3[Fe(CN)_6]$

90. Pernicious anaemia is caused by the deficiency of

(1) Vitamin A

(2) Vitamin B_1

(3) Vitamin B_{12}

(4) Vitamin B_6

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BOTANY

91. In which of the following approach, threatened animals are taken out of their natural habitat and placed in special setting where they can be protected and given special care?

(1) Wildlife sanctuaries

(2) Zoological parks

(3) National parks

(4) Biosphere reserves

92. Identify the **incorrect** statement regarding food web

(1) It is the network of interconnected food chains.

(2) Occurrence of food web provides stability to ecosystem.

(3) In a food web, a sparrow could be primary producer or secondary producer.

(4) Food webs operate because of taste preference for particular food and availability of food.

93. The rate of production of biomass or organic matter during photosynthesis is called

(1) Secondary productivity

(2) Gross primary productivity

(3) Net primary productivity

(4) Community productivity

94. Fragmentation during the decomposition is performed by

(1) Small plants

(2) Detritivores

(3) Saprophytic bacteria

(4) Saprophytic fungi

95. Humus is

(1) Non-decomposed living matter

(2) Lighter in colour

(3) Colloidal in nature and serves as a reservoir of nutrients

(4) Not decomposed in moist environment

96. Read the following statements and select the **correct** option.
Statement-A : The pyramid of energy is always upright.
Statement-B : The energy at a lower trophic level is always more than at a higher trophic level.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

97. All of the following are key criteria for determining a hotspot, **except**

(1) High degree of endemism

(2) Occurrence of particular species in all parts of the world

(3) Very high levels of species richness

(4) Degree of threat in term of habitat loss

98. Greatest biodiversity on earth is found in

(1) Amazon rain forests

(2) Temperate forest of India

(3) Western ghats of India

(4) Polar regions

99. Choose the **wrongly** matched option

(1) Nile perch – Alien species in lake Victoria

(2) Passenger Pigeon – Became extinct due to habitat fragmentation

(3) Plant pollinator mutualism – May show co-extinction

(4) *Parthenium* – Alien species in India

100. Species-area relationship curve does not explain

(1) Species richness increases with increasing explored area only upto a limit

(2) The relation between species richness and area is rectangular hyperbola for angiosperms

(3) On the log scale, relationship between species richness and area is straight line

(4) The slope of the line is much steeper for very small areas

101. Large standing crop of zooplanktons is supported by small standing crop of phytoplanktons. It represents the inverted pyramid of

(1) Energy in pond ecosystem

(2) Biomass in sea ecosystem

(3) Number in grassland ecosystem

(4) Energy for lake ecosystem

102. Read the following statements and choose the **correct** option.
Assertion (A) : GFC is major conduit for energy flow in aquatic ecosystem.
Reason (R) : DFC may be connected with GFC at some levels.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

103. Standing crop is expressed as

(1) The numbers or biomass of organisms per unit area

(2) Total amount of nutrients present in soil at any time

(3) Total number of primary producers at any time in an area

(4) Total amount of energy at any trophic level at any time

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104. Consider the given food chain.
Plant → Mice → Snake → Peacock
If 0.05 kcal of energy is available at peacock level, then how much energy was available at the plant level?

(1) 50 kcal

(2) 5000 kcal

(3) 5 kcal

(4) 500 kcal

105. Upright pyramid of number is found in all of the following ecosystems, **except**

(1) Crop field

(2) Pond

(3) Tree

(4) River

106. In India, *Parthenium*, *Lantana* and water hyacinth, all are

(1) Endangered species

(2) Critical link species

(3) Keystone species

(4) Invasive species

107. Loss of biodiversity in a region may lead to all, **except**

(1) Decline in plant productivity

(2) Reduced resistance to environmental disturbances

(3) Variability of ecosystem processes

(4) Increase in resistance to invasions by alien species

108. David Tilman performed/proposed

(1) Long term ecosystem, experiment using outdoor plot

(2) Species area relationship

(3) Integrated waste water management

(4) Rivet popper hypothesis

109. A : All animals depend on plants directly or indirectly for their food needs.
R : The consumers that feed on herbivores are called carnivores.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

110. In most ecosystems, all the pyramids of number, energy and biomass are upright. It indicates that
(a) Producers are more in number and biomass than the herbivores.
(b) Herbivores are less in number and biomass than the carnivores.
(c) Energy at a lower tropic level is always more than at a higher level.
Choose the **correct** option.

(1) Only (a) is correct

(2) Only (b) is correct

(3) Both (a) and (c) are correct

(4) Both (b) and (c) are correct

111. "In an area, a tree supports a number of herbivore birds which are eaten by one or two hawks of the area"
What will be the shape of pyramid of number and energy respectively of the above tree ecosystem?

(1) Inverted, upright

(2) Spindle, upright

(3) Spindle, inverted

(4) Upright, inverted

112. State true (T) or false (F) and choose the **correct** option
A. India has a greater ecosystem diversity than a Scandinavian country like Norway.
B. Tilman showed that in his experiments, increased diversity contributed to higher productivity.
C. Tropical amazonian rain forest is home to 40000 species of plants, 3000 species of birds and 1300 species of reptiles.

	A	B	C
(1)	T	T	T
(2)	F	T	T
(3)	T	F	F
(4)	T	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

113. Select the **correct** match.

(1) Middle lamella – Hemicellulose

(2) Algal cell wall – Galactans

(3) Primary cell wall in plants – Calcium pectate

(4) Secondary cell wall in plants – Peptidoglycans

114. "The two __A__ together with spindle fibres forms __B__."
Complete the above statement by choosing the **correct** option for A and B.

	A	B
(1)	Microtubules	Mitotic apparatus
(2)	Asters	Mitotic apparatus
(3)	Microtubules	Centriole
(4)	Asters	Centriole

(1) (1)

(2) (2)

(3) (3)

(4) (4)

115. Daughter cells formed after meiosis II are genetically

(1) Similar to each other as well as to the parent cell

(2) Dissimilar to each other but similar to the parent cell

(3) Dissimilar to each other as well as to the parent cell

(4) Similar to each other but dissimilar to the parent cell

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116. Choose the pair of cell organelles that do **not** contain any nucleic acid

(1) Plastid, mitochondria

(2) Ribosome, chloroplast

(3) Nucleus, vacuole

(4) Centriole, vacuole

117. Number of peripheral microtubules in a flagellum and centriole is respectively

(1) 18 and 27

(2) 20 and 27

(3) 9 and 27

(4) 27 and 18

118. What is/are **true** about diakinesis of prophase-I?
a. It represents transition to metaphase-I
b It is final stage of prophase-I
c. Chiasmata formation occurs
d. Nucleolus reappear

(1) Only b

(2) Only a and b

(3) Only b and c

(4) All a, b, c and d

119. Mesosomes do **not** help in

(1) Cell wall formation

(2) DNA replication

(3) Photosynthesis

(4) Respiration

120. Match **Column I** with **Column II** w.r.t stages of prophase-I.

	Column I		Column II
(A)	Leptotene	(i)	Synapsis
(B)	Zygotene	(ii)	Crossing over
(C)	Pachytene	(iii)	Desynapsis
(D)	Diplotene	(iv)	Condensation and coiling of chromatin fibres

	A	B	C	D
(1)	(ii)	(iv)	(iii)	(i)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(i)	(iii)	(iv)	(ii)
(4)	(iv)	(i)	(ii)	(iii)

(1) (1)

(2) (2)

(3) (3)

(4) (4)

121. Na^+ - K^+ pump is an example of

(1) Active transport

(2) Passive transport

(3) Transport which does not use ATP

(4) Both (2) and (3)

122. Given below diagram indicates



(1) Early prophase

(2) Metaphase

(3) Transition to metaphase

(4) Anaphase

123. Space limited by the inner membrane of the chloroplast contains

(1) Chlorophyll pigments in stroma

(2) ss circular DNA

(3) Enzymes of protein synthesis

(4) Biocatalysts for β-oxidation

124. If the amount of DNA and number of chromosome in a pollen grain are 20 pg and 16 respectively, then what was the amount of DNA and number of chromosomes in its microspore mother cell at G_2 phase?

(1) 40 pg ; 16 chromosome

(2) 80 pg ; 16 chromosome

(3) 80 pg ; 32 chromosome

(4) 40 pg ; 32 chromosome

125. The core of eukaryotic cilia and flagella is called axoneme which is made up of

(1) Filament, hook and basal body

(2) 9 centrally located doublets and 2 microtubules at periphery

(3) 2 centrally located microtubules and 9 doublets at periphery

(4) Central hub and 11 peripheral doublets

126. Read the statements and select the **correct** option.
Statement A: Both mitochondria and chloroplast are double membranous structure.
Statement B: Mitochondria contain 70S ribosomes while chloroplasts contain 80S ribosomes.

(1) Both A and B statements are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both A and B statements are incorrect

127. Tonoplast is the

(1) Single membrane around lysosome

(2) Double membrane complex around lysosome

(3) Single membrane around sap-vacuole

(4) Double membrane complex around sap-vacuole

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128. Select the **mismatched** pair

	Column-I	Column-II
(1)	Svedberg's unit	Indirect measure of density and size
(2)	Assembly of microtubules	GTP and Ca^{2+}
(3)	Prokaryotic cell	Both 70S and 80S ribosomes
(4)	Nucleus	Stained by basic dyes

(1) (1)

(2) (2)

(3) (3)

(4) (4)

129. Read the given statements (a – c) & select the **correct** ones.
(a) Schleiden reported plasma membrane as a thin outermost layer of animal cells.
(b) Unicellular organisms are capable of independent existence but do not perform all essential functions of life.
(c) In a cell, ribosomes are not found only in cytoplasm but also within and on cell organelles.

(1) (b) & (c)

(2) (a) & (c)

(3) (c) only

(4) (b) only

130. A : Anaphase-I is the actual phase where reduction in number of chromosome takes place.
R : During anaphase-I after splitting of centromere, each daughter chromatid moves to the opposite poles.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

131. State the following statements as True **(T)** or False **(F)** regarding the cell membrane and select the **correct** option.
(a) In human beings, the membrane of the erythrocyte has approximately 40 percent lipids and 52 percent proteins.
(b) Integral and peripheral proteins lie on the surface of the membrane.
(c) Quasi-fluid nature of lipids enables lateral movement of proteins within the overall bilayer.
(d) The fluid nature of the membrane helps in functions like cell growth, cell division and formation of intercellular junctions.

(1) (a)-T, (b)-T, (c)-F, (d)-T

(2) (a)-T, (b)-F, (c)-T, (d)-T

(3) (a)-T, (b)-F, (c)-F, (d)-F

(4) (a)-F, (b)-T, (c)-T, (d)-T

132. Nucleolus is the site of synthesis of ribosomal RNA which are the components of sub- units

(1) 30S, 50S

(2) 60S, 50S

(3) 40S, 60S

(4) 30S, 40S

133. Arrange the following events of meiosis in the **correct** sequence and choose the option accordingly.
(a) Formation of chiasmata
(b) Synapse formation
(c) Chromatids move towards the opposite pole
(d) Congression of chromosomes

(1) (b), (a), (d), (c)

(2) (a), (b), (d), (c)

(3) (b), (c), (d), (a)

(4) (a), (d), (b), (c)

134. Read the statements stating true **(T)** or false **(F)** and select the **correct** option.
(A) Elaioplasts store fats and oils
(B) The orange colour of carrot is due to amyloplast
(C) Stroma of chloroplast contains enzyme required for synthesis of carbohydrates and proteins.

	A	B	C
(1)	F	T	T
(2)	T	T	F
(3)	T	F	T
(4)	T	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

135. The interphase nucleus has

(1) Highly condensed and coiled DNA

(2) Highly extended and elaborate nucleoprotein fibres

(3) Always more than one nucleoli

(4) No perinuclear space

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136. How many disulphide bond(s) is/are present between A and B chains of mature insulin?

(1) One

(2) Two

(3) Three

(4) Four

137. Multi-layered epithelium that has limited role in secretion and absorption but provides protection against chemical and mechanical stresses is **not** found in

(1) Dry surface of skin

(2) Moist surface of pharynx

(3) Inner lining of pancreatic duct

(4) Inner lining of oviduct

138. A : Competent cells have more transformation efficiency then non-competitive cell.
R : Transformation efficiency is decreased by treating the cell with calcium chloride.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

139. A : A patient with ADA deficiency requires periodic infusion of own genetically engineered lymphocytes containing functional ADA cDNA.
R : The lymphocytes have a limited life span and their death again restores condition of enzyme deficiency in the patient.
In the light of above statements, select the correct option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

140. Read the following statements.
A. Diverse type of connective tissue binds together, support, strengthen and insulate other tissues.
B. Soft connective tissue includes bone and cartilage.
C. Blood is a fluid connective tissue which helps in transport.
D. Connective tissue has very less intercellular material.

	A	B	C	D
(1)	T	T	F	T
(2)	T	F	T	F
(3)	F	F	F	T
(4)	T	T	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

141. A : During electrophoresis, larger the size of DNA fragment, closer it will be to the cathode.
R : Larger fragments face more resistance as compared to smaller ones while moving through agarose matrix.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

142. Choose the **correct** statement w.r.t probe.

(1) It is always a single stranded RNA, tagged with a radioactive molecule.

(2) It is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography.

(3) It usually have the complementarity with the mutated gene.

(4) It is the single stranded DNA or RNA, not tagged with a radioactive molecule.

143. How many fragments will be generated in total upon digestion of a closed circular DNA molecule and linear DNA molecule with a same restriction enzyme having six recognition sites on both DNA molecules?

(1) 13

(2) 14

(3) 12

(4) 6

144. Read the following statements and select the **correct** option stating which ones are true and which ones are false.
a. Pro-insulin can be differentiated from mature insulin due to lack of C-chain in the former.
b. At present, about 30 recombinant therapeutics have been approved for human use worldwide.
c. Dolly was first transgenic animal.
d. Specific Bt toxin proteins were incorporated into several crop plant to make them pest resistant.

(1) a(T), b(T), c(F), d(T)

(2) a(F), b(T), c(T), d(T)

(3) a(T), b(T), c(T), d(F)

(4) a(F), b(T), c(F), d(F)

145. Simple stirred tank bioreactor can be distinguished from sparged stirred tank bioreactor based on the

(1) Absence of pH control system

(2) Presence of sampling ports

(3) Absence of sparged sterile air bubbles

(4) Presence of stirrer that reduces oxygen availability

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146. A : A tumour inducing plasmid widely used in the production of transgenic plants is that of *Agrobacterium tumefaciens*.
R : *Agrobacterium tumefaciens* is popular in genetic engineering because this bacterium can infect the roots of all cereal crops only.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

147. Read the following statements and select the **correct** option.
I. RNAi involves post-translational silencing of a specific mRNA.
II. Silencing of mRNA occurs by formation of ds RNA molecule thus preventing transcription of parasitic mRNA.

(1) Both statements are correct.

(2) Statement I is correct but II is incorrect.

(3) Statement II is correct but I is incorrect.

(4) Both statements are incorrect.

148. Consider the following statements.
(a) The intercellular material is solid and pliable, resisting compression
(b) Cells are enclosed in small cavities called lacunae within the matrix secreted by them
(c) The tissue interacts with smooth muscles to bring about body movements
(d) It is the site of production of blood cells
The statements which are applicable to cartilage include

(1) (a), (b) and (d)

(2) (a), (c) and (d)

(3) (c) and (d)

(4) (a) and (b)

149. In the following question, a statement of assertion (A) is followed by a statement of reason (R).
A : Bt cotton is resistant to cotton bollworm.
R : Toxin from *Bacillus thuringiensis* was introduced into cotton plants.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

150. Genetic modification in plants has

(1) Made crops susceptible to abiotic stresses

(2) Increased reliance on chemical pesticides

(3) Increased efficiency of mineral usage by plants

(4) Helped to increase post harvest losses

151. Select the **incorrect** match w.r.t. a typical agarose gel electrophoresis.

(1)	Ethidium bromide	–	Bright orange bands
(2)	DNA fragments	–	Move towards negative electrode
(3)	Agarose	–	Extracted from sea weeds
(4)	Elution	–	Extraction of desired DNA fragments from the agarose gel

(1) (1)

(2) (2)

(3) (3)

(4) (4)

152. Which of the following sequences can be recognised by a restriction enzyme?

(1) 5' GAATTC 3'
5' GTTAAG 3'

(2) 5' GATC 3'
3' CTAG 5'

(3) 5' GTTA 3'
3' CAAT 5'

(4) 5' CATGG 3'
5' GTACC 3'

153. Read statements A and B and select the **correct** option.
Statement-A : DNA is a hydrophilic molecule, thus it cannot pass through cell membrane.
Statement-B : Electroporation creates transient microscopic pores in cell membrane making entry of DNA molecules into the cell easier.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct but B is incorrect

(4) Statement A is incorrect but B is correct

154. Which organisation in India deals with the safety of introducing GM-organisms for public services?

(1) CDRI

(2) GEAC

(3) WTO

(4) NIH

155. Gene *cry II Ab* of *B. thuringiensis* forms toxin for the control of

(1) Corn borer

(2) Cotton bollworms

(3) *M. incognita*

(4) Baculoviruses

156. Non-transformant *E.coli* are those which have not received any foreign DNA. These non-transformants

(1) Grow on nutrient broth

(2) Grow on nutrient broth with ampicillin

(3) Grow on nutrient broth with tetracycline

(4) Grow on nutrient broth with tetracycline and ampicillin

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157. *Cry* endotoxins obtained from *Bacillus thuringiensis* are not effective against

(1) Mosquitoes

(2) Flies

(3) Nematodes

(4) Bollworms

158. The four sketches (A, B, C and D) given below represent four different types of animal tissues. Which of these is **incorrectly** identified in the options given, along with its location and function?

(A)



(B)



(C)



(D)



	Tissue	Location	Functions
(1) (A)	Simple squamous epithelium	Air sacs of lungs	Form the diffusion boundary
(2) (B)	Simple cuboidal epithelium	Tubular parts of nephrons	Secretion and absorption
(3) (C)	Simple columnar ciliated epithelium	Fallopian tube	Move the particles or mucus in a specific direction over the epithelium
(4) (D)	Connective tissue	Between adjacent bones of the vertebral column	Site of production of blood cells

(1) (1)

(2) (2)

(3) (3)

(4) (4)

159. How many cycles of PCR are required to obtain 32 copies from a single piece of dsDNA?

(1) 20

(2) 5

(3) 10

(4) 15

160. Gene of interest was cloned at site *Hind III* in pBR322 by a researcher and then transformed into *E. coli*. The recombinant plasmid will exhibit

(1) Susceptibility to tetracycline

(2) Resistance to both ampicillin and tetracycline

(3) Susceptibility to ampicillin

(4) Resistance to tetracycline only

161. Which of the following does **not** show presence of connective tissue fibres of collagen and elastin?

(1) Bone

(2) Blood

(3) Cartilage

(4) Tendon

162. What is the term used for inserting a healthy copy of a gene into a person who has a defective gene?

(1) Cloning vector

(2) Gene therapy

(3) Recombinant DNA

(4) Polymerase chain reaction (PCR)

163. How many of the following can be classified under upstream and downstream processing respectively?

Expression of protein in desired host, Separation of desired product, Purification of desired product, Isolation of DNA from host cell, Making multiple copies of gene

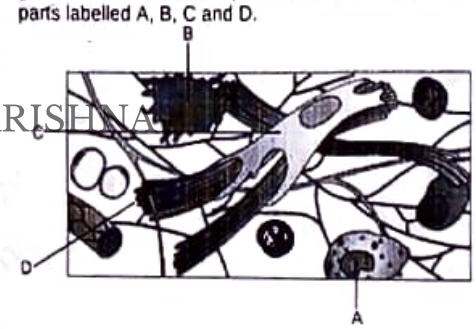
(1) 3, 2

(2) 2, 3

(3) 2, 1

(4) 1, 1

164. Diagrammatic sketch of a certain connective tissue is given below. Select the option that **correctly** identifies the parts labelled A, B, C and D.



	A	B	C	D
(1)	Adipocyte	Mast cell	Collagen fibres	Macrophage
(2)	Collagen fibres	Osteocyte	Mast cell	Fibroblast
(3)	Macrophage	Mast cell	collagen fibres	Chondrocyte
(4)	Mast cell	Macrophage	Fibroblast	Collagen fibres

(1) (1)

(2) (2)

(3) (3)

(4) (4)

165. Select the **incorrect** pair w.r.t. following neuroglial cells and their functions

(1) Astrocytes – Formation of blood brain barrier

(2) Oligodendrocytes – Formation of myelin sheath in CNS

(3) Microglial cells – Phagocytosis

(4) Schwann cells – Form neurilemma in CNS neurons

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166. The main challenge for production of insulin using rDNA technique was

(1) Its modification after translation by the addition of sugar molecules

(2) Getting insulin assembled into a mature form with optimal activity

(3) Transformation of this proinsulin into host *E.coli*

(4) Screening of transformed *E.coli* by blue-white selection

167. A PCR experiment can be completed in a few hours whereas it takes weeks to obtain gene by cloning. Despite of many advantages of PCR, gene cloning is used why?

(1) PCR cannot be used to isolate the gene that has not been studied before

(2) There is a limit to the length of DNA sequence that can be amplified by PCR

(3) It is difficult to synthesise specific primers if the sequence of nucleotides of DNA segment of foreign gene are not known

(4) All of these

168. Genes are altered by manipulation and reinserted into an individual of same species then this individual can be called

(1) Genetically modified organism

(2) Hybrid organism

(3) Recombinant organism

(4) More than one option is correct

169. What happens to alien piece of DNA linked to origin of replication of plasmid, which is transferred into new organism?

(1) The transferred DNA will always mutate

(2) The rDNA will replicate autonomously itself in the progeny cells of the organism, independent of host

(3) The DNA will always associate with the host cell genome at any circumstance

(4) It will replicate rapidly and the alien cell will be killed by lysis always

170. The ciliated epithelial cells are mainly present in

(1) Bronchioles and fallopian tube

(2) Eustachian tube and salivary duct

(3) Fallopian tube and pancreatic duct

(4) Bronchioles and mammary duct

171. Why is the gene coding for Bt toxin called 'cry'?

(1) *cry* gene codes for crystalline protein

(2) *cry* gene exist as crystal

(3) *cry* gene codes for cryophilic protein

(4) *cry* gene codes for Bt toxin

172. Which type of neurons are most common in the adult human body and are found in brain and spinal cord?

(1) Unipolar neurons

(2) Multipolar neurons

(3) Bipolar neurons

(4) Pseudounipolar neurons

173. Complete the analogy by choosing the correct option.
Annealing: 50–60°C:: Extension: _____.

(1) 94°C

(2) 40°C

(3) 72°C

(4) 89°C

174. Plasmids, A, B and C with different molecular size are being investigated through agarose gel electrophoresis.
'A' is a linearised plasmid with 7000 bp.
'B' is a circular plasmid with 16000 bp.
'C' is a supercoiled plasmid with 6000 bp.
Which of the following is **true**?

(1) 'A' travels the fastest towards the anode

(2) 'B' travels the fastest towards the cathode

(3) 'C' will be closest to anode after electrophoresis

(4) Both 'A' and 'C' will migrate simultaneously towards cathode

175. Intercalated disc is characteristic feature of cardiac muscle fibres. These intercalated discs function as boosters of cardiac impulse due to presence

(1) Communication junctions

(2) Gap junctions

(3) Tight junctions

(4) Both (1) and (2)

176. Find out the **correct** statement w.r.t epithelial tissues.

(1) Basement membrane of epithelial tissues is a cellular layer

(2) Glands are composed of epithelial tissues

(3) There is an abundant intercellular material between the epithelial cells

(4) Simple epithelium is present in Buccal cavity, pharynx and in urinary bladder

177. A : A researcher finds that recombinant colonies obtained are white in color.
R : He concludes that insertional inactivation of gene β-galactosidase has taken place.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

178. Read the following statements carefully
Statement-A : The native (wild type) *E.coli* cells do not carry resistance against any antibiotics
Statement-B : Lac Z gene in pBR322 is considered as useful selectable marker for *E.coli*
Choose the **correct** option

(1) Both statements are incorrect

(2) Both statements are correct

(3) Only statement A is correct

(4) Only statement B is correct

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179. Select the **mismatch**.

(1) Flavr Savr – Transgenic potato

(2) Gene therapy – ADA

(3) Tracy – α-1 antitrypsin

(4) Humulin – *E.coli*

180. Select the option that includes characteristics applicable to plasmids

a. Linear DNA.
b. Circular DNA.
c. Present in bacteria and some yeast cells.
d. Contain essential genes / vital genes.
e. Extra chromosomal self replicating DNA.

(1) Only a & e

(2) Only b, c & e

(3) Only b, c, d & e

(4) Only a

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